How do we model predictive uncertainty?



To get into more details of the modelling of those two types of aleatoric uncertainty, it is important to understand Maximum Likelihood (ML) and Maximum a Posteriori (MAP) estimation of unknown variables (e.g., model parameters)



Now, knowing about MLE, MAP, and Bayesian Modelling, how do we model aleatoric or epistemic uncertainty?



How do we model predictive uncertainty?

How do we model aleatoric uncertainty?

$$y_{true} = y + noise$$

Let's call noise ϵ, now does it depend on the input $\mathbf{x}$?

Homoscedastic: $y_{true} = y + \epsilon$

Heteroscedastic: $y_{true} = y + \epsilon(\mathbf{x})$

Assume a distribution for noise → (e.g., normal)

$y_{true} = y + \mathcal{N}(0, \sigma^2)$

$y_{true} = y + \mathcal{N}(0, \sigma(\mathbf{x})^2)$

Then insert/estimate/learn the parameters of the *assumed* noise distribution



How do we model predictive uncertainty?

modeling for aleatoric uncertainty

1) Have the model f_θ output a distribution over y.

2) Find θ with MLE.

Maximum Likelihood estimate of an expressive-enough model f_θ on a large-enough dataset can perfectly capture aleatoric uncertainty. Why?



How do we model predictive uncertainty?

modeling for aleatoric uncertainty (known/estimated case)

1) Assume some noise distribution. e.g., in the normal noise:

Homoscedastic: $y_{true} = y + \epsilon$

Heteroscedastic: $y_{true} = y + \epsilon(\mathbf{x})$

Assume a distribution for noise → (e.g., normal)

$y_{true} = y + \mathcal{N}(0, \sigma^2)$

$y_{true} = y + \mathcal{N}(0, \sigma(\mathbf{x})^2)$

2) Then, estimate its parameters. e.g., $\sigma^2 = a^2$

3) Then, have the loss accordingly.

$$l(f_{\theta}(\mathbf{x}_i), y_i) = -\log \mathcal{N}(f_{\theta}(\mathbf{x}_i), \sigma^2 = a^2)\Big|_{y_i} = -\log\left(\frac{1}{\sqrt{2\pi a^2}} e^{\frac{-(f_{\theta}(\mathbf{x}_i)-y_i)^2}{2a^2}}\right) = \frac{\log a^2}{2} + \frac{(f_{\theta}(\mathbf{x}_i) - y_i)^2}{2a^2} + C$$

MSE loss



note that variance linearly discounts the MSE

How do we model predictive uncertainty?

modeling for aleatoric uncertainty (learnable case)

1) Assume some noise distribution.

Homoscedastic: $y_{true} = y + \epsilon$

Heteroscedastic: $y_{true} = y + \epsilon(\mathbf{x})$

Assume a distribution for noise → (e.g., normal)

$y_{true} = y + \mathcal{N}(0, \sigma^2)$

$y_{true} = y + \mathcal{N}(0, \sigma(\mathbf{x})^2)$

2) Then, have the model output distributions $\mathcal{N}\left(f_{\boldsymbol{\theta}}(\mathbf{x}_i), f'_{\boldsymbol{\theta}'}\right)$ following the noise distribution.

3) Then, modify the loss accordingly.

e.g., in the homoscedastic normal noise:

$$l(f_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = -\log \mathcal{N}\left(f_{\boldsymbol{\theta}}(\mathbf{x}_i), f'_{\boldsymbol{\theta}'}\right)\Big|_{y_i} = -\log\left(\frac{1}{\sqrt{2\pi f'_{\boldsymbol{\theta}'}}} e^{\frac{-(f_{\boldsymbol{\theta}}(\mathbf{x}_i)-y_i)^2}{2f'_{\boldsymbol{\theta}'}}}\right) = \frac{\log f'_{\boldsymbol{\theta}'}}{2} + \frac{(f_{\boldsymbol{\theta}}(\mathbf{x}_i) - y_i)^2}{2f'_{\boldsymbol{\theta}'}} + C$$

Learnt Variance logarithmically increases the loss

MSE loss!

note that ***learnt*** variance linearly discounts the MSE

We'll see the heteroscedastic case later

How do we model predictive uncertainty?

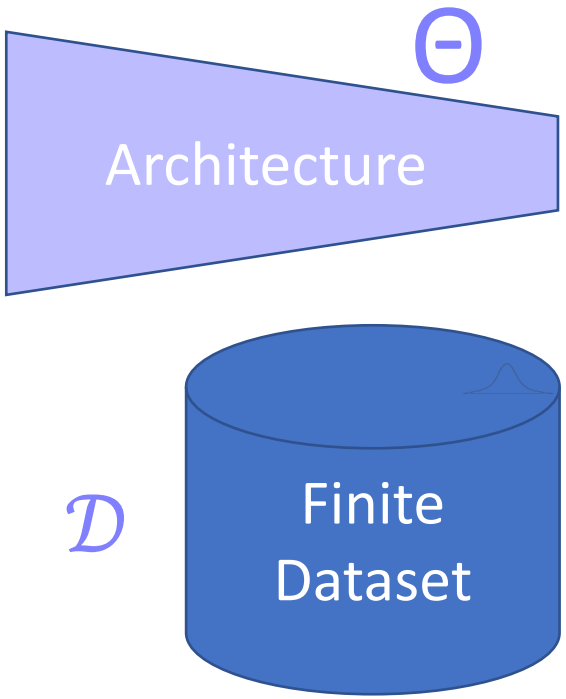
Modeling for epistemic uncertainty – the general approach

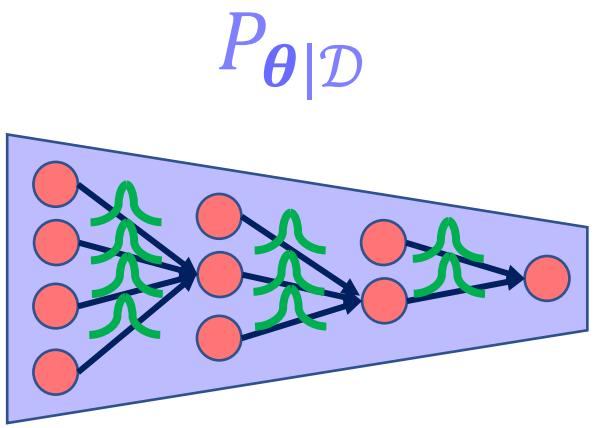
- Acknowledge uncertainty in model parameters (e.g., network weights)
- Model that uncertainty in model parameters
- Have a mechanism to transfer such uncertainty into the prediction



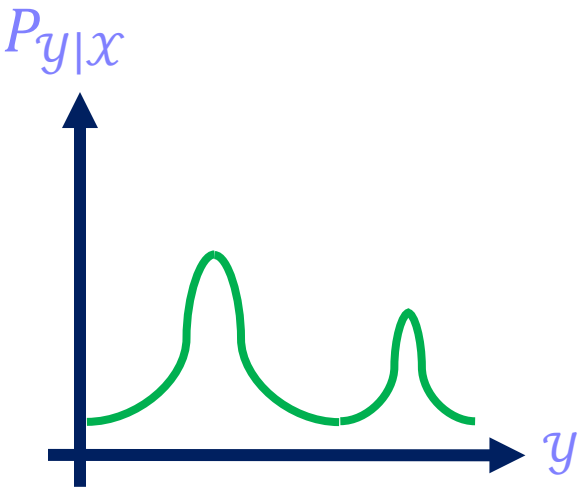
How do we model predictive uncertainty?

Modeling for epistemic uncertainty – the general picture





learn uncertainty of model parameters



transfer it to
uncertainty in model predictions

How do we model predictive uncertainty?

How do we model epistemic uncertainty?

$(P(\boldsymbol{\theta}|\mathcal{D}))$



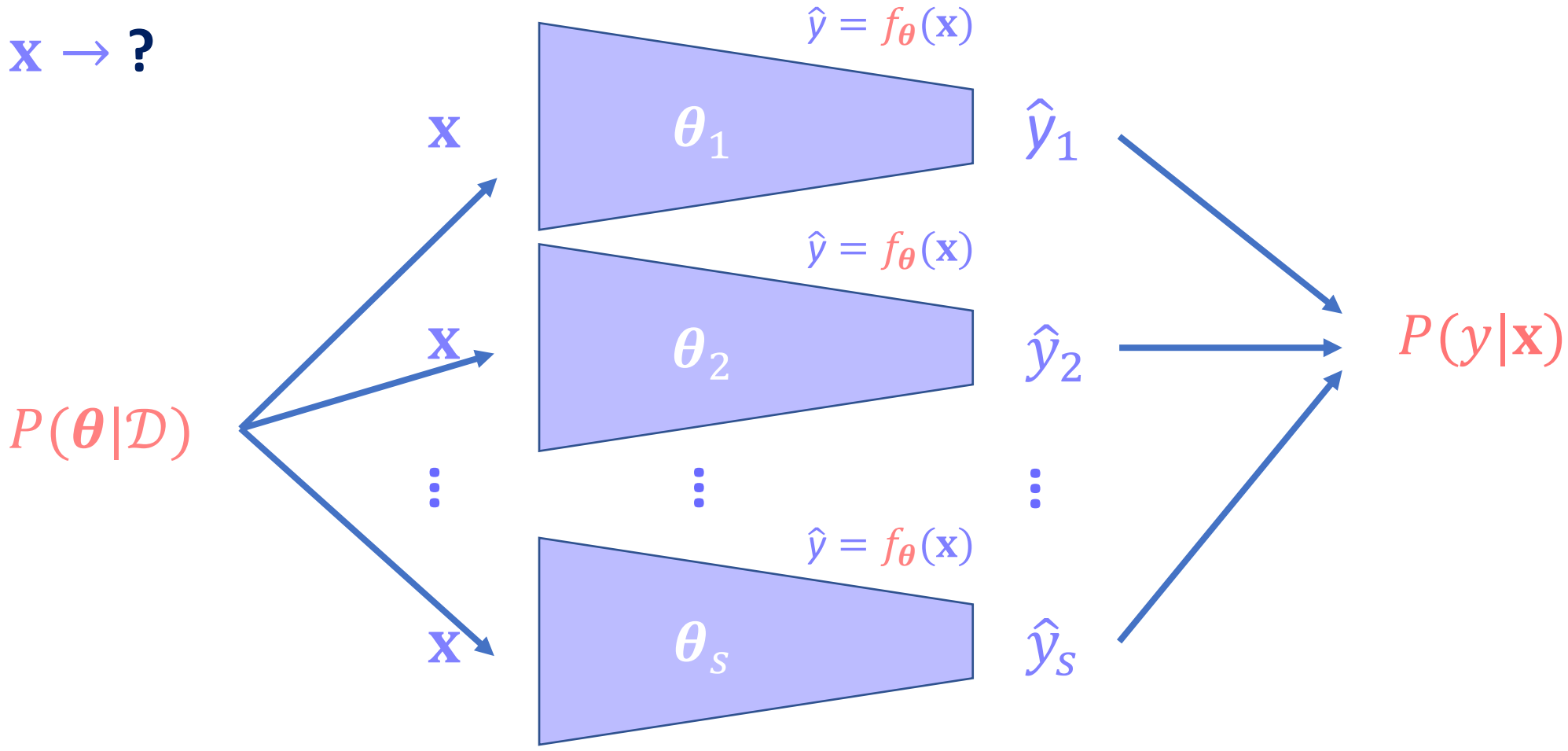
- Bayesian Modelling

- Non-Bayesian (frequentist) Modelling
  - ensemble methods
  - conformal prediction
  - etc.



How do we model predictive uncertainty?

***Frequentist* modeling for epistemic uncertainty – a simple intuition**



How do we model predictive uncertainty?

Bayesian modeling for epistemic uncertainty – the formal definition

$$\boldsymbol{\theta}^* = argmax_{\boldsymbol{\theta}} P(\mathcal{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})$$

Parameter estimation: $$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{P(\mathcal{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\mathcal{D})}$$

- We want the full distribution to be able to transfer the uncertainty in model parameters (due to lack of knowledge) to uncertainty in prediction

How do we model predictive uncertainty?

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- We want the full distribution to be able to transfer the uncertainty in model parameters (due to lack of knowledge) to uncertainty in prediction
- We usually would need the Bayesian evidence (P(D)) to be able to have normalized predictive distribution as follows:

Prediction: $$P(y|\mathbf{x},\mathcal{D}) = \int_{\Theta} P(y,\boldsymbol{\theta}|\mathbf{x},\mathcal{D})d\boldsymbol{\theta} = \int_{\Theta} P(y|\mathbf{x},\mathcal{D},\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathbf{x},\mathcal{D})d\boldsymbol{\theta} = \int_{\Theta} P(y|\mathbf{x},\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}$$

How do we model predictive uncertainty?

Bayesian modeling for epistemic uncertainty – the formal steps

for discriminative learning:

- As in MAP:
  - We assume a prior $P(\boldsymbol{\theta})$
  - We devise a likelihood function $P(y|\mathbf{x}, \boldsymbol{\theta})$
- Assuming $\mathcal{D}$ is *i.i.d.*, we infer the full posterior over parameters $\boldsymbol{\theta}$, $P(\boldsymbol{\theta}|\mathcal{D})$: $$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{\prod_i P(y_i|\mathbf{x}_i, \boldsymbol{\theta})\, P(\boldsymbol{\theta})}{P(\mathcal{D})}$$
- At test time, for a new datapoint: $$P(y|\mathbf{x}, \mathcal{D}) = \int_{\Theta} P(y|\mathbf{x}, \boldsymbol{\theta}) P(\boldsymbol{\theta}|\mathcal{D}) d\boldsymbol{\theta}$$